

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \Upsilon_1 k^2 & -\Upsilon_1 k^2 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\Upsilon_1 k^2 & \Upsilon_1 k^2 \\ \hline \end{array} \quad \mathcal{K}_{1^-}^{\#1} \quad \mathcal{K}_{1^-}^{\#2} \\
\mathcal{K}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{\Upsilon_2 k^2}{3} & -\frac{1}{3} \sqrt{5} \Upsilon_3 k^2 \\ \hline \end{array} \\
\mathcal{K}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{1}{3} \sqrt{5} \Upsilon_3 k^2 & -\frac{\Upsilon_4 k^2}{3} \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1} \alpha\beta \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{K}_{3^-}^{\#1} \alpha\beta\chi \\
\mathcal{K}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -2 \Upsilon_1 k^2 \\ \hline \end{array}
\end{array}$$

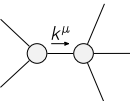
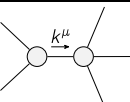
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{1}{4 k^2 \binom{(4)}{\kappa_1 + \kappa_2}} & -\frac{1}{4 k^2 \binom{(4)}{\kappa_1 + \kappa_2}} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{1}{4 k^2 \binom{(4)}{\kappa_1 + \kappa_2}} & \frac{1}{4 k^2 \binom{(4)}{\kappa_1 + \kappa_2}} \\ \hline \end{array} \quad \mathcal{J}_{1^-}^{\#1} \quad \mathcal{J}_{1^-}^{\#2} \\
\mathcal{J}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{\Upsilon_4 k^2}{3 \text{Det}(1^-)} & \frac{\sqrt{5} \Upsilon_3 k^2}{3 \text{Det}(1^-)} \\ \hline \end{array} \\
\mathcal{J}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{\sqrt{5} \Upsilon_3 k^2}{3 \text{Det}(1^-)} & \frac{\Upsilon_2 k^2}{3 \text{Det}(1^-)} \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1} \alpha\beta \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{J}_{3^-}^{\#1} \alpha\beta\chi \\
\mathcal{J}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{1}{2 k^2 \binom{(4)}{\kappa_1 + \kappa_2}} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &= \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \& \& \Upsilon_2 = 2 \binom{(4)}{\kappa_1} + 3 \binom{(4)}{\kappa_2} \& \& \Upsilon_3 = 2 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \& \& \\
\Upsilon_4 &= 6 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \& \& \text{Det}(1^-) = -\frac{8}{9} k^4 \binom{(4)}{\kappa_1 + \kappa_2} (4 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2})
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}_\chi^\delta \partial^\chi \mathcal{K}_\alpha^{\beta\delta} + \binom{(4)}{\kappa_2} \partial_\chi \mathcal{K}_\beta^\delta \partial^\chi \mathcal{K}_\alpha^{\beta\delta} + 6 \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + 6 \binom{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\
& 4 \binom{(4)}{\kappa_1} \partial^\chi \mathcal{K}_\alpha^{\beta\delta} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 4 \binom{(4)}{\kappa_2} \partial^\chi \mathcal{K}_\alpha^{\beta\delta} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \binom{(4)}{\kappa_1} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \binom{(4)}{\kappa_2} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#2} + 3 \mathcal{J}_{0^+}^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha{}^\beta = 0$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$	5	$k^\chi (\partial_\epsilon \partial_\chi \partial^\beta \partial^\alpha \mathcal{J}_\delta^{\delta\epsilon} + 2 \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\delta\epsilon} - 3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta\delta\epsilon} -$ $3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\alpha\delta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta} \partial_\phi \partial_\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta} \partial_\phi \partial^\phi \partial_\epsilon \partial_\chi \mathcal{J}_\delta^{\delta\epsilon}) = 0$	
Total # constraint(s):	6		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{50 \binom{(4)}{\kappa_1} + 35 \binom{(4)}{\kappa_2} - \sqrt{196 \binom{(4)}{\kappa_1}^2 + 620 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_2} + 649 \binom{(4)}{\kappa_2}^2}}{4 \binom{(4)}{\kappa_1}^2 + 5 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_2}^2}$
	2	0	$\frac{50 \binom{(4)}{\kappa_1} + 35 \binom{(4)}{\kappa_2} + \sqrt{196 \binom{(4)}{\kappa_1}^2 + 620 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_2} + 649 \binom{(4)}{\kappa_2}^2}}{4 \binom{(4)}{\kappa_1}^2 + 5 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_2}^2}$
Resolved unitarity condition(s):		$(\binom{(4)}{\kappa_2} \leq 0 \& \& \binom{(4)}{\kappa_1} > -\binom{(4)}{\kappa_2}) \parallel (\binom{(4)}{\kappa_2} > 0 \& \& \binom{(4)}{\kappa_1} > -\frac{\binom{(4)}{\kappa_2}}{4})$	